```
Parameters:
  WhitelistedIpAddress:                                    ①
    Description: 'Whitelisted IP address'
    Type: String
    AllowedPattern: '^[0-9]{1,3}\.[0-9]{1,3}\.[0-9]{1,3}\.[0-9]{1,3}$'
    ConstraintDescription: 'Enter a valid IPv4 address'
Resources:
  SecurityGroup:
    Type: 'AWS::EC2::SecurityGroup'
    Properties:
      GroupDescription: 'Learn how to protect your EC2 Instance.'
      VpcId: !Ref VPC
      # [...]
      SecurityGroupIngress:
      # [...]
      - Description: 'allowing inbound HTTP traffic'
        IpProtocol: tcp
        FromPort: '80'
        ToPort: '80'
        CidrIp: !Sub '${WhitelistedIpAddress}/32'          ②
```

① Public IP address parameter

② Uses WhitelistedIpAddress/32 as a value to turn the IP address input into a CIDR

Update the CloudFormation stack with the template located at https://s3.amazonaws.com/awsinaction-code3/chapter05/firewall4.yaml. When asked for parameters, type in your public IP address for `WhitelistedIpAddress`. Now only your IP address can open HTTP connections to your EC2 instance.

Classless Inter-Domain Routing (CIDR)

You may wonder what `/32` means. To understand what's going on, you need to switch your brain into binary mode. An IP address is 4 bytes or 32 bits long. The `/32` defines how many bits (32, in this case) should be used to form a range of addresses. If you want to define the exact IP address that is allowed, you must use all 32 bits.

Sometimes it makes sense to define a range of allowed IP addresses. For example, you can use `10.0.0.0/8` to create a range between 10.0.0.0 and 10.255.255.255,